

- 1 The rate of heat flow, R , can be found using the equation

$$R = k A \frac{(T_2 - T_1)}{L}$$

where k is the thermal conductivity, A is the total cross sectional area of the conducting surface, and L is the thickness of conducting surface separating the 2 temperatures T_1 and T_2 .

What is the SI base units for k ?

- A** $\text{kg m s}^{-3} \text{K}^{-1}$ **B** $\text{kg m s}^{-2} \text{K}^{-1}$ **C** kg m s^{-3} **D** $\text{kg s}^{-2} \text{K}^{-1}$

Ans: A

$$\begin{aligned} [k] &= \left[\frac{R}{A (T_2 - T_1)} \right] \\ &= \frac{\text{W m}}{\text{m}^2 \text{K}} \\ &= \text{W m}^{-1} \text{K}^{-1} = \text{J s}^{-1} \text{m}^{-1} \text{K}^{-1} = \text{kg m s}^{-3} \text{K}^{-1} \end{aligned}$$

- 2 With reference to Question 1, a student collected the following measurements to determine the rate of heat flow R of the object

$$L = (0.35 \pm 0.05) \text{ mm}$$

$$T_1 = (32.0 \pm 0.5) ^\circ\text{C}$$

$$T_2 = (4.0 \pm 0.5) ^\circ\text{C}$$